Dynamic OSM-Based Profile Routing from Public GPS Trace Signals

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Abstract

Personalized route recommendation is usually evaluated with clean navigation logs, explicit user histories, or proprietary trajectory data. This paper advances a narrower open-data alternative: preference-aware route-candidate reranking from OpenStreetMap (OSM) features and public GPS-derived movement signals. We present a working system that generates OSM route candidates, extracts interpretable route features, constructs contextual pseudo-history profiles, and reranks candidates using profile, prompt, hybrid, or trajectory-derived vehicle profile scoring. The paper makes validity an explicit part of the method by evaluating trace reconstruction and route-candidate reranking separately, rather than treating public GPS traces as unquestioned user histories. On an exploratory Toronto OSM public-trace probe, timestamp coverage and GPS-to-route proximity are strong, while route-distance ratio identifies where reconstruction quality must still improve. The primary public benchmark path is a 100-query Porto taxi trajectory evaluation with OSM driving-route candidates, feature/path oracles, bootstrap confidence intervals, a learned feature-ranker baseline, paired bootstrap comparisons, and NASR-style path-overlap metrics. In the current Porto benchmark, a trajectory-derived vehicle profile improves ranking metrics over shortest-distance and a generic profile, especially Hit@3 and MRR, while a supervised prior-query feature ranker provides a stronger internal learned baseline on Hit@3 and mean feature distance. Path-level metrics still expose limitations: shortest-distance remains competitive on path F1/NDTW, and oracles show that map matching and candidate generation shape the achievable ceiling. The result is initial public benchmark evidence for a transparent, reproducible, interpretable OSM route-candidate reranking framework under realistic open-data constraints.

CCS Concepts

• **Information systems** → **Location based services**; **Recommender systems**; • **Computing methodologies** → *Natural language processing*.

Keywords

route-candidate reranking, OpenStreetMap, GPS traces, public trajectory benchmark, pseudo-history profiles

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1 Introduction

Route choice is not only a shortest-path problem. A person may prefer a greener, quieter, simpler, better-lit, or more familiar route even when that route is not the minimum-distance option. This makes routing a natural personalization problem: the best route depends on the origin and destination, but also on the request context and the traveler's implicit or explicit preferences. Prior work on personalized route recommendation often assumes large-scale navigation logs, clean historical route choices, or trained models over dense trajectory datasets [2, 3, 6]. This paper targets a complementary open-data setting: interpretable route-candidate reranking that can be built, inspected, and reproduced with OSM features and public GPS-derived movement signals.

This paper studies the following question: can OSM-derived historical movement signals support preference-aware route-candidate reranking when clean user histories are unavailable? We use the term *pseudo-history profile* because public GPS traces provide movement evidence, not guaranteed persistent user identity. That identity gap is the core research challenge. The system therefore treats trace reconstruction quality as a measured component of the pipeline and evaluates route-candidate reranking separately from trace reconstruction.

We present a working OSM-based route-candidate reranking system. Given an origin, destination, request context, optional preference text, and optional history records, the backend generates OSM route candidates, extracts interpretable route features, constructs dynamic profile weights from contextual history, and reranks candidates using profile, prompt, or hybrid modes. The implementation is exposed through a FastAPI backend and React interface, preserving the product-facing demonstration that motivated the project. This matters because route selection is ultimately an interactive decision aid: the system must produce ranked alternatives and explanations that a user can inspect, not only offline scores.

The evaluation is designed around this validity challenge. A first OSM public-trace experiment demonstrates the end-to-end pipeline for Toronto walking traces and uses route-distance ratio to expose reconstruction cases that require improvement. This Toronto probe is exploratory internal validation, not the headline public benchmark. The primary public benchmark is Porto: it reconstructs observed OSM driving routes from the public Porto taxi trajectory dataset, generates route candidates for the same origin and destination, and reports ranking metrics plus path-overlap metrics with bootstrap confidence intervals. Walking OSM traces and Porto taxi trajectories represent different travel modes, so they are not merged into one headline result. The Porto setup creates a concrete bridge from an open-data reranking system to future same-data comparisons against learned route-search methods such as NASR [6].

The contributions are:

- An end-to-end OSM route-candidate reranking architecture that combines candidate generation, interpretable route features, contextual pseudo-history profiles, prompt-based reranking, and a GUI/API workflow.

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- A trace-to-profile pipeline that converts public GPS-derived signals into pseudo-history records while reporting reconstruction quality instead of assuming clean user histories.
- A route-candidate evaluation protocol in which generated candidates are ranked against held-out observed routes, avoiding history-record ranking artifacts that can overstate shortest-distance baselines.
- A Porto public-dataset benchmark with random, shortest-distance, profile, trajectory-derived vehicle profile, learned feature-ranker, feature-oracle, and path-oracle comparisons using Hit@K, MRR, NDCG@3, feature distance, path F1, NDTW, edit distance, paired bootstrap differences, and confidence intervals.

The central claim is that profile-aware OSM route-candidate reranking can be evaluated transparently against non-personalized baselines in an open-data setting, and that the current Porto benchmark shows measurable ranking gains for trajectory-derived profiles while path metrics expose remaining limitations. The remaining research frontier is to strengthen map matching, candidate generation, and reproduced external comparisons so that the same framework can support a direct benchmark claim.

2 Related Work

2.1 Personalized Route Recommendation

Personalized route recommendation has been studied with stronger data assumptions than this work can currently make. Dai et al. use large-scale trajectory data to model personalized route recommendation from historical movement behavior [2]. More recent user-habit work models route sorting from historical navigation behavior and compares against minimum-ETA, LightGBM, and DCN-v2 baselines with inconsistency-oriented metrics [3]. These papers motivate our use of route features and historical profiles, but they assume cleaner user or vehicle histories than OSM public trace signals provide.

Neural search methods frame route recommendation as an origin-destination search problem. NASR integrates neural models with A* search for personalized route recommendation and evaluates route recovery with path-level metrics such as precision, recall, F1, and edit distance [6]. This is the strongest top-tier methodological reference for our next benchmark target because it affects route generation itself rather than only post-hoc reranking. Our current system is simpler: it generates a small set of OSM candidates and reranks them with interpretable profile or prompt signals. The Porto benchmark is designed to move the evaluation closer to NASR-style route recovery without pretending that a full NASR reproduction has already been completed.

NASR learns the search process itself, while our system reranks candidates generated by OSM graph search. Therefore NASR is a stronger route-generation/search method, and our current comparison is not a direct reproduction. Our contribution is interpretable reranking and diagnostic evaluation under open-data constraints. A direct NASR comparison requires reproducing its preprocessing, splits, and path-overlap metrics on the same dataset.

2.2 Trajectory Reconstruction and Historical Route Planning

Trajectory reconstruction and popular-route mining are directly relevant because our pseudo-history profiles depend on reconstructed traces. Chen et al. study popular-route construction from uncertain trajectories and use trajectory similarity metrics such as normalized dynamic time warping and maximum distance [1]. Tian et al. propose DRPK for effective route planning from historical trajectories on road networks [5]. These works motivate a strict separation between reconstruction quality and ranking quality. A route ranker can appear to perform well if the reconstructed reference path is poor or if the candidate pool is artificially easy; therefore, we report route-distance ratio, GPS-to-route distance, path F1, NDTW, edit distance, and oracle upper bounds.

2.3 Contextual and Preference-Aware Recommendation

Contextual route recommendation datasets and route-choice studies show that origin-destination routing is affected by context, not only graph distance [4]. Preference-driven travel-planning systems such as Personal Travel Solver use LLM and solver components to infer user preferences and produce personalized travel plans [7]. Our work is narrower. We do not claim full itinerary planning or language-grounded route generation. Instead, natural-language preferences are optional inputs used to rerank already-generated OSM route candidates. When preference text is synthetic, we label it as synthetic and treat prompt/SBERT results as ablations rather than evidence of real stated preferences.

2.4 Positioning

The closest direct future comparison is not the OSM public-trace experiment alone. The defensible comparison path is a same-data public benchmark, most likely Porto, where our candidate reranker and external route-recovery baselines are evaluated with the same path-overlap metrics. Until that reproduction is complete, this paper positions the proposed method as an interpretable, open-data route-candidate reranking framework with transparent diagnostics, rather than as a system that already surpasses learned personalized search methods.

Among the related systems, NASR is the closest reproducible top-tier route-recovery benchmark because it uses origin-destination trajectory recovery and path-overlap metrics; the 2024 user-habit route recommendation paper is the closest conceptual match for profile-based route sorting, but it is not directly comparable without its private navigation logs and route-choice labels.

3 Problem Statement

Let a route-candidate reranking request be defined by origin o, destination d, request context c, optional preference text p, and optional historical records H_u for a user or pseudo-user. The system must generate a candidate set $R = \{r_1, \ldots, r_k\}$ and return an ordering that places routes aligned with observed or inferred preferences near the top.

Each route r_i is represented by an interpretable feature vector:

Table 1: Baseline alignment. The table separates conceptual relevance from direct comparability so that the evaluation does not force an invalid comparison.

Work	Main task	Data signal	Comparable metrics	Role in this paper
User-habit route recommendation [3]	Personalized vehicle route sorting	Historical navigation logs	Inconsistency-oriented ranking metrics	Closest conceptual profile-ranking reference, but not directly comparable without clean user logs.
NASR [6]	Personalized route search/recovery	Trajectory data and road graphs	Precision, recall, F1, edit distance	Main future technical benchmark target on public trajectory data.
Big trajectory PRR [2]	Driver-specific route recommendation	Large GPS trajectory data	Route-cost and preference-oriented evaluation	Older but important driver-preference baseline family.
RICK [1]	Popular route inference from uncertain trajectories	Sparse/noisy trajectories	NDTW, maximum distance	Reconstruction-quality reference, not a personalized ranking baseline.
DRPK [5]	Realistic route planning from historical trajectories	Large historical trajectories	Path prediction and efficiency metrics	Non-personalized historical-trajectory planning reference.
Context Trails / PTS [4, 7]	Contextual route/travel recommendation	Contextual POI trails or reviews	Recommendation and preference-satisfaction metrics	Motivates context and preference modeling; task differs from OSM route reranking.

$$x_i = [\text{distance}, \text{major}, \text{walk}, \text{residential}, \text{service}, \text{turns}, \ldots]. \tag{1}$$

For OSM-derived pseudo-history evaluation, the held-out record is not treated as a generated candidate. Instead, it is treated as an observed reference route with feature vector x^*. The oracle generated candidate is:

$$r^* = \arg\min_{r_i \in R} \text{dist}(\hat{x}_i, \hat{x}^*), \tag{2}$$

where $\hat{x}$ denotes min-max normalized features within the evaluation set.

The research question is:

> Can dynamic profiles built from OSM-derived historical movement signals rerank generated OSM route candidates closer to observed pseudo-route behavior than non-personalized baselines?

This is intentionally narrower than claiming full personalized navigation. The paper evaluates a reranking problem: candidate routes are already generated by OSM graph search, and the proposed profile module changes their order. This distinction matters because a learned search method such as NASR can generate different paths, while our current implementation can only choose among candidates in the generated pool.

3.1 Evaluation Assumptions

The evaluation uses two kinds of observed reference routes. In the OSM public-trace setting, references are reconstructed walking-route pseudo-histories from public GPS trackpoints. In the Porto setting, references are reconstructed driving routes from public taxi trajectories. These are different travel modes and are not merged into one headline result. In both cases, the observed reference is imperfect. We therefore report reconstruction diagnostics and oracle upper bounds rather than treating the held-out route as unquestioned ground truth.

3.2 Baseline Objective

For each held-out query, a method returns an ordering of generated candidates. A successful method should rank the oracle candidate near the top, where the oracle is either the candidate closest in normalized route-feature space or, for path diagnostics, the candidate with strongest overlap with the reconstructed reference path. The main deployable baselines are random, shortest-distance, profile, and trajectory-derived vehicle profile ranking. Feature and path oracles are diagnostic upper bounds, not usable methods.

4 Methodology

The system contains six main stages: trace processing, approximate map matching, feature extraction, profile construction, candidate generation, and route-candidate reranking. Figure 1 summarizes the data flow.

4.1 OSM Trace Processing

The pipeline probes OSM public GPS trackpoints in a Toronto bounding box, parses GPX files, and builds temporal/spatial pseudo-segments. Segments split when timestamp gaps or spatial jumps exceed configured thresholds. Useful segments are retained when they have sufficient distance and plausible speed. These segments

OSM public GPS signals / Porto taxi trajectories → temporal-spatial segmentation → approximate map matching → route feature extraction → dynamic profile construction → OSM candidate generation → reranking and evaluation

Figure 1: End-to-end framework. Public movement signals are converted into reconstructed route references and pseudo-history profiles. For a new origin-destination query, the system generates OSM candidates and reranks them with baseline, profile, prompt, or hybrid methods.

are explicitly treated as OSM-derived historical movement signals, not clean per-user histories.

4.2 Approximate Map Matching

For each useful pseudo-segment, the system builds a local OSM walking graph, maps GPS points to nearest graph nodes, deduplicates consecutive repeated nodes, and connects successive matched nodes with shortest paths. This approximate map matching is intentionally simple and therefore reported with diagnostics: matched-point coverage, pair success rate, GPS-to-route distance, and route-distance ratio.

For Porto taxi trajectories, the evaluator uses an OSM driving graph and simplifies dense GPS traces into spatially separated anchors before shortest-path stitching. The anchor spacing parameter reduces redundant point-to-point stitching when consecutive samples fall on nearby road nodes. The current benchmark uses adaptive anchor simplification: several deterministic anchor spacings are tried, and the reconstruction with the best route-distance-ratio and GPS-fit tradeoff is selected. The default setting starts from a 120 meter minimum anchor spacing and a 50-anchor cap, which reduces route-distance-ratio inflation while preserving enough trajectory structure for path-overlap evaluation.

4.3 Route Feature Extraction

For both reconstructed pseudo-history routes and generated candidate routes, the system extracts OSM-derived features: route distance, major-road percentage, walking-way percentage, residential percentage, service-road percentage, intersections, turns, park proximity, lighting percentage, traffic signals, crossings, tunnel length, and an OSM-only safety proxy. The safety proxy is not a crime model; it is a route-attribute score based on available OSM tags.

4.4 Dynamic Profile Construction

Given a request timestamp, the profile module computes temporal context: weekday/weekend, time bucket, season, and rush-hour status. It selects history records with similar context, averages route features, and converts those averages into feature weights. Context rules then adjust weights for rush hour, evening/night, weekend, summer, and winter.

For the Porto benchmark, the generic walking profile is supplemented with a trajectory-derived vehicle profile. This profile uses earlier Porto trips as contextual movement history and emphasizes driving-relevant features: distance, major-road exposure, residential/service-road share, turns, intersections, traffic signals, and tunnel length. The method remains unsupervised: it does not train on held-out labels, it does not assume stated human preferences, and it does not use the oracle candidate during scoring.

4.5 Candidate Generation

The route generator constructs a local OSM graph around the query and returns a small set of route alternatives. The walking pipeline uses OSM walking networks and route attributes such as footways, crossings, lighting, and park proximity. The Porto pipeline uses a shared OSM driving graph around central Porto and generates up to ten diverse vehicle candidates by interleaving multiple edge-weight variants, including shortest, major-road-preferring, major-road-avoiding, calmer local-road, low-intersection, and balanced driving costs. Keeping the candidate pool small preserves runtime for the GUI/API workflow and makes the reranking problem inspectable. The reported feature and path oracles quantify the upper bound imposed by this candidate pool.

4.6 Reranking Modes

The system supports the following evaluation methods:

- **Random:** random permutation of generated route candidates.
- **Shortest-distance:** candidates sorted by OSM route length.
- **Profile:** candidates scored using dynamic profile weights.
- **Prompt/SBERT:** route summaries reranked by semantic similarity to preference text.
- **Hybrid:** 0.75 profile score plus 0.25 prompt/SBERT score, matching the backend.
- **Learned feature ranker:** a random-forest regressor trained only on earlier query candidates to predict feature distance to the reconstructed reference.

Prompt/SBERT and hybrid modes are skipped when no preference text exists. This is the current case for the Porto public trajectory benchmark.

4.7 Scoring

Profile scoring compares each candidate feature vector with a contextual target profile. For numeric feature j, the score rewards closeness between candidate value x_{ij} and target value μ_j under profile weight w_j:

$$s_{\text{profile}}(r_i) = -\sum_j w_j \cdot |\hat{x}_{ij} - \hat{\mu}_j|. \tag{3}$$

The prompt/SBERT mode converts each route to a textual summary and scores semantic similarity against preference text. The hybrid mode combines normalized profile and prompt scores:

$$s_{\text{hybrid}}(r_i) = 0.75\, s_{\text{profile}}(r_i) + 0.25\, s_{\text{prompt}}(r_i). \tag{4}$$

These weights match the backend implementation and are treated as fixed system settings, not tuned hyperparameters.

5 Evaluation

The evaluation has three layers: reconstruction quality, threshold sensitivity, and route-candidate reranking quality. This separation is intentional because public OSM GPS trackpoints are not clean per-user histories.

5.1 Reconstruction Quality

The trace pipeline reports timestamp coverage, pseudo-segments found, useful segment rate, map-match success rate, GPS-to-route distance, and route-distance ratio. Strict usability requires timestamp coverage at least 0.80, at least five useful segments, map-match success rate at least 0.70, median GPS-to-route distance at most 100 meters, and median route-distance ratio in $[0.5, 2.0]$. Exploratory usability relaxes only the route-distance ratio ceiling to 2.5.

5.2 Threshold Sensitivity

To avoid tuning a single threshold setting, we evaluate five segmentation/filtering configurations: the current configuration, a stricter time-gap split, a stricter spatial-jump split, a stricter minimum-distance filter, and a lower speed ceiling. Each configuration is evaluated with the same reconstruction diagnostics. This study tests whether the OSM-derived pseudo-history pipeline is stable under reasonable threshold changes.

5.3 Route-Candidate Baselines

The route-candidate reranking evaluation uses generated route candidates. For each held-out pseudo-history record, the evaluator:

(1) Uses earlier records as profile history.
(2) Uses the held-out origin and destination.
(3) Generates up to $k = 5$ OSM walking-route candidates for the OSM trace setting, or up to $k = 10$ OSM driving candidates for the Porto setting.
(4) Treats the held-out reconstructed route features and coordinates as the observed reference.
(5) Selects the oracle candidate by minimum normalized feature distance to the observed reference.
(6) Ranks candidates using each available baseline.

This replaces a history-record ranking sanity check in which the held-out record was ranked against previous history records. That older setup made shortest-distance look artificially strong because the held-out pseudo-route could be the shortest candidate by construction.

5.4 Baselines

We compare random ranking, shortest-distance ranking, profile ranking, prompt/SBERT ranking, and hybrid ranking. For Porto, we additionally report a trajectory-derived vehicle profile ranker, a learned feature ranker, a feature-oracle upper bound, and a path-oracle upper bound. The trajectory-derived vehicle profile ranker uses earlier Porto trip features but emphasizes driving-relevant signals such as distance, road-class mix, turns, intersections, and traffic signals. The learned feature ranker is trained only on prior query candidates and their feature-distance labels, so it is an internal learned baseline rather than an external reproduced method. The feature oracle ranks by the same feature distance used to select the ranking oracle. The path oracle ranks by path overlap with the reconstructed route. Both oracle rows are used only to diagnose candidate-pool quality. Prompt/SBERT and hybrid use synthetic preference text in the OSM pseudo-history experiment; this text is labeled as synthetic and is used only as an ablation, not as evidence of real stated user preferences.

5.5 Metrics

Ranking quality is measured with Hit@1, Hit@3, MRR, NDCG@3, and mean feature distance. Hit@K is one when the oracle candidate appears in the first K positions. MRR is the reciprocal rank of the oracle candidate averaged across queries. NDCG@3 discounts lower-ranked relevant candidates and gives a graded top-three ranking view. Mean feature distance reports how close the top-ranked candidate is to the held-out reference in normalized feature space.

Path similarity is measured with point-overlap precision/recall/F1, normalized dynamic time warping (NDTW), coordinate edit distance, and maximum nearest-point distance between the top-ranked route and the held-out reconstructed route. For the Porto public-dataset benchmark, we also report 95% bootstrap confidence intervals over queries.

5.6 External Benchmark Goal

The current OSM pseudo-history evaluation is an internal open-trace evaluation, while the Porto benchmark is the primary public-data path for external comparison. The Porto setup uses public taxi trajectories, simplifies dense GPS samples into spatially separated reconstruction anchors, reconstructs an observed route on an OSM driving graph, generates OSM route candidates for the same origin and destination, and reports ranking metrics plus path-overlap metrics. This creates the evaluation substrate needed for NASR-style route recovery comparisons with matched data, splits, and metrics. We do not report NASR as reproduced in this paper because the official implementation requires its own preprocessing and train/test format; reproducing it under the same Porto split remains the strongest next external-comparison step.

6 Results

This section reports the current system outputs and benchmark evidence. The results demonstrate a functioning end-to-end framework, initial gains for profile-aware route-candidate reranking in Porto, and clear diagnostics for the remaining comparison gap to reproduced external systems.

6.1 Trace Reconstruction

The Toronto OSM trace probe currently contains 25,000 parsed trackpoints with timestamp coverage 1.00. The pipeline found 11 pseudo-segments and 11 useful segments, for a useful segment rate of 1.00. Approximate map matching succeeded for all attempted useful segments. The median GPS-to-route median distance is approximately 6.99 meters, indicating close geometric alignment between sampled points and reconstructed routes. However, the median route-distance ratio is approximately 2.37, above the strict target ceiling of 2.0.

Therefore, the current strict reconstruction flag is not satisfied, while the exploratory usability flag is satisfied under the looser 2.5 route-distance-ratio ceiling. This distinction is useful: the data supports pseudo-history ranking experiments, and the quality report identifies exactly where route reconstruction must improve for stronger route-fidelity claims.

6.2 Threshold Sensitivity

Table 2 shows that the route-distance-ratio finding is stable across several reasonable threshold choices. This evidence shows the reconstruction diagnosis is not an artifact of one chosen configuration.

6.3 Route-Candidate Ranking

The corrected route-candidate baseline evaluation runs on 8 held-out pseudo-history queries. Table 3 reports the ranking metrics.

The profile method improves over shortest-distance on Hit@1, MRR, NDCG@3, path F1, NDTW, edit distance, and maximum path distance in the current run. Because this OSM public-trace set is small and consists of walking-style public traces, we use it as end-to-end validation rather than the headline external comparison. The larger role of this result is to show that the profile-reranking mechanism changes route ordering in the intended direction under a corrected candidate-ranking protocol.

6.4 Path-Similarity Metrics

Against held-out reconstructed route coordinates, the profile method achieves mean path F1 of 0.630 and mean NDTW of 0.351, compared with 0.601 and 0.344 for shortest-distance. These path metrics provide an interpretable check that ranking improvements are not only feature-space effects.

6.5 Porto Public-Dataset Benchmark

For the primary same-data public benchmark, we ran a Porto taxi trajectory experiment using the public ECML/PKDD taxi trajectory data. This experiment samples trips from the Porto training file, reconstructs an observed OSM driving route using adaptive GPS anchors, generates up to ten diverse OSM driving-route candidates for the same origin and destination, and reranks the candidates using random, shortest-distance, generic profile, trajectory-derived vehicle profile, and learned feature-ranker baselines. The learned ranker starts after enough prior candidate-label examples exist, so it is evaluated on 97 of the 100 successful queries. We also report two diagnostic, non-deployable oracle upper bounds: a feature oracle, which ranks candidates by the same feature-distance criterion used to define the ranking oracle, and a path oracle, which ranks candidates by path overlap with the reconstructed route. Prompt/SBERT and hybrid methods are skipped because Porto trajectories do not include natural-language route preferences.

In this 100-query run, the trajectory-derived vehicle profile improves over shortest-distance and the generic profile on Hit@1, Hit@3, MRR, NDCG@3, and mean feature distance. Paired bootstrap differences strengthen this interpretation but also narrow it: vehicle profile versus shortest-distance is significant for Hit@3 and MRR, but not for path F1 or NDTW; vehicle profile versus generic profile is significant for Hit@1, Hit@3, MRR, NDCG@3, and feature distance, but again not for path F1/NDTW. The learned feature ranker, trained only on prior query candidates, is competitive with the vehicle profile and obtains the best deployable Hit@3 and mean feature distance; paired differences over shortest-distance are significant for Hit@3 and MRR. Adaptive-anchor reconstruction reduces the median observed route-distance ratio to 1.22 and the mean ratio to 1.57, with median GPS-to-route distance of 26.84 meters; 20 of 100 evaluated queries remain above route-distance ratio 2.0. The path oracle reaches 0.658 path F1 and 0.427 NDTW, improving the candidate-pool ceiling relative to the earlier five-candidate setup. At the same time, shortest-distance remains strongest among deployable baselines on top-ranked path F1/NDTW, so the result should be read as a ranking-metric gain with a clearly identified path-quality bottleneck, not as a solved route-recovery result.

We attempted to scale the Porto benchmark to 250 successful queries with the same graph and candidate-generation settings. That run did not finish within a 10-minute local execution budget, so the paper reports the completed 100-query result and treats larger-scale Porto evaluation as future work rather than fabricating an extrapolated result.

6.6 Result Summary

The OSM public-trace experiment shows that the full pseudo-history pipeline is operational and quality-instrumented. The Porto experiment provides the stronger public-dataset evaluation path: trajectory-derived vehicle profile reranking improves over shortest-distance and the generic profile on ranking metrics, the learned feature ranker supplies a stronger internal learned baseline, and path metrics/oracle ceilings identify the next technical leverage point. Because Toronto walking traces and Porto taxi trajectories are different modes, the two experiments should be read as complementary diagnostics rather than one combined headline result.

7 Discussion

The current system is strongest as a research framework: it exposes the full chain from OSM public GPS signals to route reconstruction, feature extraction, profile construction, candidate generation, route-candidate reranking, and evaluation. The GUI/demo angle remains valuable because it shows how a user would interact with ranked routes and profile explanations.

The main validity consideration is data identity. OSM public GPS trackpoints cannot be assumed to represent clean, persistent user histories. The current pseudo-history profiles are useful for studying whether trace-derived movement signals can inform route ranking, but they are not equivalent to logged user route choices. This motivates the paper's explicit terminology and separate reconstruction diagnostics.

The second validity consideration is reconstruction quality. GPS-to-route distance is favorable, but route-distance ratio is high for several segments. The threshold-sensitivity study shows this diagnosis persists across reasonable segmentation and filtering choices. The Porto benchmark now uses simplified GPS anchors before shortest-path stitching, reducing route-distance-ratio inflation and

Table 2: Threshold sensitivity over OSM-derived pseudo-history reconstruction. All configurations remain exploratory-usable but fail strict usability due to route-distance ratio.

Config	Useful	GPS med.	Ratio med.	Strict
Current	11	6.99	2.37	No
Stricter gap	12	6.72	2.36	No
Stricter jump	11	6.99	2.37	No
Stricter distance	10	7.41	2.36	No
Lower speed ceiling	9	6.32	2.35	No

Table 3: Route-candidate baseline results over eight held-out OSM pseudo-history queries. Prompt/SBERT and hybrid use synthetic preference text and are reported as ablations.

Method	Hit@1	Hit@3	MRR	NDCG@3	Mean Dist.
Random	0.375	0.750	0.583	0.668	2.622
Shortest-distance	0.250	0.625	0.456	0.690	2.579
Profile	0.375	0.625	0.540	0.727	2.582
Prompt/SBERT	0.250	0.500	0.481	0.673	2.552
Hybrid	0.250	0.500	0.467	0.672	2.613

Table 4: Porto public-dataset route-candidate benchmark over 100 successful sampled queries. Values in parentheses are 95% bootstrap confidence intervals.

Method	Hit@1	Hit@3	MRR
Feature oracle (diag.)	1.000 (1.000, 1.000)	1.000 (1.000, 1.000)	1.000 (1.000, 1.000)
Path oracle (diag.)	1.000 (1.000, 1.000)	1.000 (1.000, 1.000)	1.000 (1.000, 1.000)
Random	0.120 (0.060, 0.180)	0.330 (0.240, 0.430)	0.306 (0.257, 0.364)
Shortest-distance	0.120 (0.060, 0.190)	0.290 (0.200, 0.370)	0.300 (0.249, 0.354)
Profile	0.100 (0.050, 0.170)	0.210 (0.130, 0.290)	0.264 (0.211, 0.319)
Traj.-vehicle profile	0.230 (0.150, 0.320)	0.430 (0.330, 0.530)	0.408 (0.343, 0.475)
Learned feature ranker	0.216 (0.144, 0.309)	0.454 (0.361, 0.557)	0.401 (0.331, 0.470)

Method	NDCG@3	Mean Dist.	Path F1	NDTW
Feature oracle (diag.)	0.980 (0.950, 1.000)	1.378 (1.226, 1.526)	0.575 (0.506, 0.645)	0.367 (0.300, 0.439)
Path oracle (diag.)	1.000 (1.000, 1.000)	1.569 (1.407, 1.722)	0.658 (0.596, 0.715)	0.427 (0.356, 0.499)
Random	0.536 (0.492, 0.581)	1.858 (1.735, 1.976)	0.483 (0.426, 0.542)	0.257 (0.199, 0.313)
Shortest-distance	0.545 (0.482, 0.607)	1.819 (1.674, 1.951)	0.512 (0.450, 0.571)	0.290 (0.229, 0.351)
Profile	0.445 (0.383, 0.508)	1.977 (1.860, 2.077)	0.445 (0.388, 0.506)	0.215 (0.162, 0.270)
Traj.-vehicle profile	0.615 (0.561, 0.672)	1.819 (1.700, 1.922)	0.472 (0.415, 0.523)	0.240 (0.183, 0.302)
Learned feature ranker	0.597 (0.533, 0.652)	1.725 (1.583, 1.865)	0.477 (0.416, 0.540)	0.269 (0.203, 0.339)

making the reconstruction process more stable. The next improvement is a stronger map-matching stage so strict route-fidelity thresholds can be met more often.

The third validity consideration is evaluation size and external comparability. The current route-candidate evaluation uses eight held-out queries from one OSM public trace area and 100 successful sampled Porto queries. This establishes the benchmark machinery and public-data comparison path; reproducing an external baseline under matched Porto splits is the next step toward a direct paper-to-paper superiority claim.

The main path to a strong paper is a same-data public benchmark. NASR-style evaluation on a public dataset such as Porto taxi trajectories would let us compare route-recovery or route-ranking methods with recognized metrics such as precision, recall, F1, and edit distance. A direct claim of superiority should only be made after matching the dataset, task definition, train/test split, and metrics of an external baseline or a faithful reproduction.

The Porto benchmark is the main evidence path. It shows that the trajectory-derived vehicle profile scorer can be evaluated on public trajectory data and improves over shortest-distance and the generic profile on several ranking metrics. The learned feature ranker provides a stronger internal learned comparator, while the diverse candidate generator improves the path-oracle ceiling relative to the earlier five-candidate setup. Random and shortest-distance remain competitive on some top-k and path metrics, which is useful evidence about candidate-pool difficulty rather than a failure of the framework. The diagnostic oracle rows show where path-level quality is constrained by candidate generation and reconstructed route references, pointing to a clear next research target.

Prompt/SBERT and hybrid results currently use synthetic preference text generated from reconstructed route features. These are useful as ablations because they test the prompt-ranking machinery, but they do not prove performance with real user-stated preferences. If prompt or hybrid personalization becomes a central claim, the system needs collected preference text or a controlled preference study.

7.1 Role of Language Models

GPT/LLM components are not the core ranking model in the current evaluation. The deployed research pipeline is OSM graph construction, route reconstruction, route feature extraction, profile scoring, candidate generation, and route-candidate reranking. SBERT/prompt mode is an optional semantic reranking ablation when preference text exists, and Porto results do not depend on real user-stated prompts. Future work could use GPT-style models for preference summarization or a natural-language interface, but the present contribution should not be overstated as GraphRAG unless an explicit knowledge-graph retrieval component is added.

7.2 Threats to Validity

The most important internal-validity threat is oracle definition. Feature-oracle ranking rewards closeness to reconstructed route features, which may favor methods that match aggregate properties rather than exact geometry. Path-oracle metrics partly address this, but they remain bounded by reconstruction quality. The most important external-validity threat is that walking-style OSM pseudo-histories and Porto taxi trajectories represent different travel modes. We therefore do not combine them into one headline result. The most important construct-validity threat is interpreting pseudo-history profiles as user preferences. The paper avoids that interpretation unless clean user identity or stated preferences are available.

7.3 Path to a Stronger Claim

The strongest publishable claim should be tested on a larger same-data benchmark. A defensible target is: given public Porto trajectories and OSM road networks, compare profile-aware candidate reranking against shortest-path, random, learned feature ranking, and a reproduced NASR-style route-recovery baseline using the same path-overlap metrics. If the profile-aware method improves ranking metrics while maintaining competitive path F1/NDTW and the GUI/API remains usable, the project can claim an interpretable, open-data route-ranking alternative. The current paper establishes the system, evaluation protocol, and first public-dataset evidence needed to support that trajectory.

8 Reproducibility

The implementation is organized as a FastAPI backend and React frontend. The backend exposes route-candidate reranking through the same workflow used by the evaluation scripts: graph construction, route generation, feature extraction, profile construction, reranking, and result serialization. The OSM pseudo-history files, quality reports, baseline outputs, and Porto benchmark outputs are written under the project data directory. The public Porto CSV is intentionally not committed because of its size; the evaluator writes a status report if the dataset is missing.

The main OSM route-candidate evaluation can be reproduced with the route-candidate baseline script. The public Porto benchmark can be reproduced after placing the Porto training CSV at `data/porto/train.csv`. The 100-query run reported in this paper uses:

```
python scripts\evaluate_porto_candidate_baselines.py
  --max-tests 100 --max-rows-to-scan 30000
  --sample-stride 20 --k-routes 10
  --dist-meters 2500 --bootstrap-samples 1000
  --min-anchor-spacing-m 120 --max-anchors 50
  --shared-graph-radius-m 8000
```

The script outputs both JSON and CSV summaries, including skipped-baseline reasons. Prompt/SBERT and hybrid are skipped for Porto because the public trajectories do not contain natural-language preferences. This behavior is deliberate: missing data should produce a reported skip, not fabricated preference labels.

Paired bootstrap differences and the ablation summary can be regenerated from the Porto JSON artifact with:

```
python scripts\summarize_porto_paired_differences.py
```

This writes paired-bootstrap and ablation summaries under `data/`. The no-temporal trajectory-derived vehicle profile ablation requires rerunning the updated Porto evaluator; it is not backfilled into earlier artifacts.

9 Conclusion

This paper presented a dynamic OSM-based route-candidate reranking system that builds pseudo-history profiles from public GPS-derived movement signals, extracts interpretable OSM route features, and reranks generated route candidates using profile, prompt, or hybrid modes. The main methodological point is separation: reconstruction quality, candidate-generation quality, and reranking quality are evaluated independently so that noisy public traces are not mistaken for clean user histories.

The current results show a clear research path without claiming final superiority over learned route-search systems. In the Toronto OSM public-trace setting, the pipeline produces usable exploratory pseudo-history signals and reports reconstruction diagnostics that expose route-fidelity gaps. In the primary Porto public-dataset benchmark, a trajectory-derived vehicle profile improves over shortest-distance and a generic profile on ranking metrics, while path metrics identify map matching and candidate generation as the next technical leverage points. The result is a reproducible research framework with a concrete public-benchmark agenda: larger-scale Porto evaluation, reproduced external baselines such as NASR under matched preprocessing and splits, and stronger map matching can now be added without changing the system architecture or evaluation philosophy.

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